GREENFIELD DEVELOPERS PVT LTD
BIM Coordination — Request for Information

RFI #RFI-PRJ-2024-GT-001-CLH-005

Issued: 21 September 2026
Auto-generated by BIM Clash Engine

● CRITICAL SEVERITY | HARD CLASH | ELECTRICAL vs MEP | BASEMENT B1 | GRID C2 / D2

PROJECT INFORMATION

Project Name Greenfield Tower — Block A	**Project Number** PRJ-2024-GT-001	**Client** Greenfield Developers Pvt Ltd
Location Sector 62, Noida, Uttar Pradesh	**RFI Number** RFI-PRJ-2024-GT-001-CLH-005	**Date Issued** 21 September 2026

RFI SUBJECT

RFI-CLH-005: Critical Hard Clash Resolution - 11kV HV Cable Tray vs AHU Fresh Air Duct at Basement B1 Grid C2/D2

CLASH ELEMENT DETAILS

Field	Element A — Electrical	Element B — MEP
Revit ID	REV-EL-014332	REV-MEP-008891
Category	Cable Tray	HVAC Duct
Family	Ladder Tray 600mm wide	Rectangular Duct 1000x600
Size	600mm x 100mm	1000mm x 600mm
System	HV Distribution — 11kV Feeder	—
Level	Basement B1	Basement B1
Grid Ref	C2 / D2	C2 / D2
Clash Point	X: 33.670 Y: 9.880 Z: 2.400	Penetration: -0.220m

TECHNICAL DESCRIPTION

A hard clash exists between the 600mm wide 11kV HV Ladder Cable Tray (REV-EL-014332) and the 1000x600mm AHU Fresh Air Duct (REV-MEP-008891) at Basement Level B1, Grid C2/D2, with a penetration depth of -0.220m. The current design violates the mandatory 500mm clear separation requirement for HV services as per IS 1646 and NBC 2016, creating an immediate safety and compliance hazard.

IMPACT ASSESSMENT

Failure to resolve this critical clash will result in non-compliance with electrical safety codes, potential project hold points from the statutory authority, and significant rework costs if not addressed before fabrication and installation of the HVAC ductwork.

APPLICABLE CODES & STANDARDS

- IS 1646:2012 (Code of Practice for Electrical Installations in Buildings)
- NBC 2016 (National Building Code of India) - Part 10: Electrical Installations
- OISD 12 (Oil Industry Standard Data) - Electrical Safety Guidelines
- IS 732 (Code of Practice for Low Current Electrical Installations) - for general coordination context

PROPOSED RESOLUTION

Redesign the HVAC duct routing at Grid C2/D2 to maintain a minimum vertical or horizontal separation of 500mm from the 11kV HV cable tray. The AHU fresh air duct should be rerouted to pass above or below the cable tray with adequate clearance, or the cable tray support structure should be adjusted to allow the duct to pass through a dedicated clearance zone, ensuring no physical contact and maintaining the required safety distance.

Option 1: Reroute the 1000x600mm AHU fresh air duct to pass above the 11kV cable tray with a minimum vertical clearance of 500mm, adjusting the duct hanger points accordingly.
Option 2: Reroute the 1000x600mm AHU fresh air duct to pass below the 11kV cable tray, ensuring the duct top is at least 500mm below the bottom of the cable tray, and verify structural support capacity for the duct at the new elevation.
Option 3: Install a fire-rated separation barrier (if applicable and approved by the fire consultant) between the HV cable tray and the HVAC duct, provided the physical separation can be reduced to a minimum of 250mm with the barrier, subject to strict approval from the Electrical Engineer and Fire Consultant.

INFORMATION REQUIRED

Please provide a revised HVAC duct routing plan for Grid C2/D2 at Basement B1 that demonstrates compliance with the 500mm separation requirement from the 11kV HV cable tray. Include updated Revit model elements and a clash-free coordination drawing for approval.

ROUTING & RESPONSE DEADLINE

Assigned To	Role	Email	Response Required
Rajesh Kumar	Lead MEP Engineer	devendragupta56789@gmail.com	24 hours

Issued By: Devendra Gupta | BIM Automation Engineer
Company: Greenfield BIM Team | **Date:** 21 September 2026 | **Auto-generated:** 21-Sep-2026 09:27